

NATIONAL JUNIOR COLLEGE, SINGAPORE
Senior High 2
Preliminary Examination
Higher 2

CANDIDATE
NAME

BIOLOGY
CLASS

REGISTRATION
NUMBER

BIOLOGY

Paper 2

9648/02

30 August 2013

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your Biology class, registration number and name on all the work you hand in.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

The number of marks is given in brackets [] at the end of each question or part question.

Section A

Write your answers in this booklet using dark blue or black pen.

Section B

Write your answers on the separate Answer Paper using dark blue or black pen.

For Examiner's Use	
Paper 2	
Section A	
1	/ 11
2	/ 15
3	/ 10
4	/ 12
5	/ 10
6	/ 12
7	/ 10
Section B	
8 / 9	/ 20
TOTAL	/ 100

This document consists of **20** printed pages.

[Turn over

Section A

Answer all the questions in this section.

- 1 Fig. 1.1 is a transmission electron micrograph of a mammalian cell.

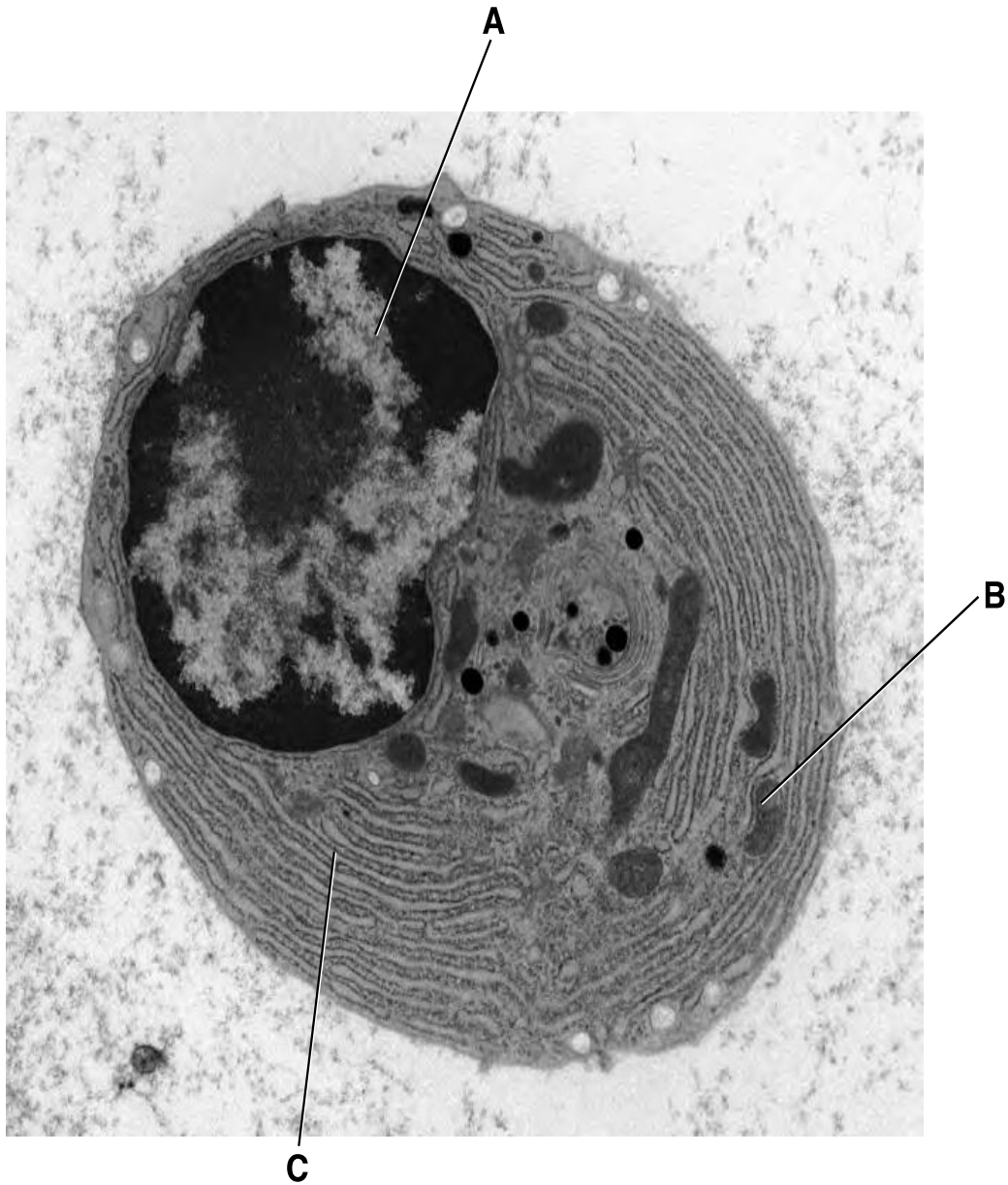


Fig. 1.1

- (a) Complete Table 1.1 to:

- name in full, structures **A**, **B**, and **C**.
- outline how each structure functions to contribute to its specific role in the mammalian cell.

Table 1.1

<i>structure</i>	<i>name of structure</i>	<i>function of structure within mammalian cell</i>
A		
B		
C		

[3]

Table 1.2 below represents the ionic concentrations inside and outside a typical mammalian cell.

Table 1.2

	<i>normal conditions</i>	
	<i>concentration (mM)</i>	
<i>ions</i>	<i>intracellular</i>	<i>extracellular</i>
Na ⁺	5-15	145
H ⁺	7×10^{-5}	4×10^{-5}
Cl ⁻	5-15	110

(b) Explain how the intracellular concentration of Na⁺ is maintained.

[1]

-
-
-
-
-
-
-
-
- [4]

- (i) Suggest two ways in which tetracycline acts at ribosomes to inhibit protein synthesis.

[2]

- [1]

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- 2 Meselson and Stahl performed the following experiment with the bacterium *Escherichia coli* (*E. coli*). They grew cells for several generations in the presence of the heavy isotope of nitrogen, ^{15}N . All the DNA of the progeny that were produced from the initial small inoculum was labelled with ^{15}N . The ^{15}N -labelled cells were then placed in ^{14}N -containing medium. The DNA produced in subsequent generations was extracted. The DNA from the different generations was compared using the technique of differential centrifugation using a caesium chloride density gradient to separate the DNA containing ^{15}N from DNA containing ^{14}N , as ^{15}N labelled DNA is more dense than ordinary ^{14}N DNA. The position of the ^{15}N and ^{14}N DNA absorbance was determined in ultraviolet light at 260 nm. The results obtained are shown in Fig. 2.1.

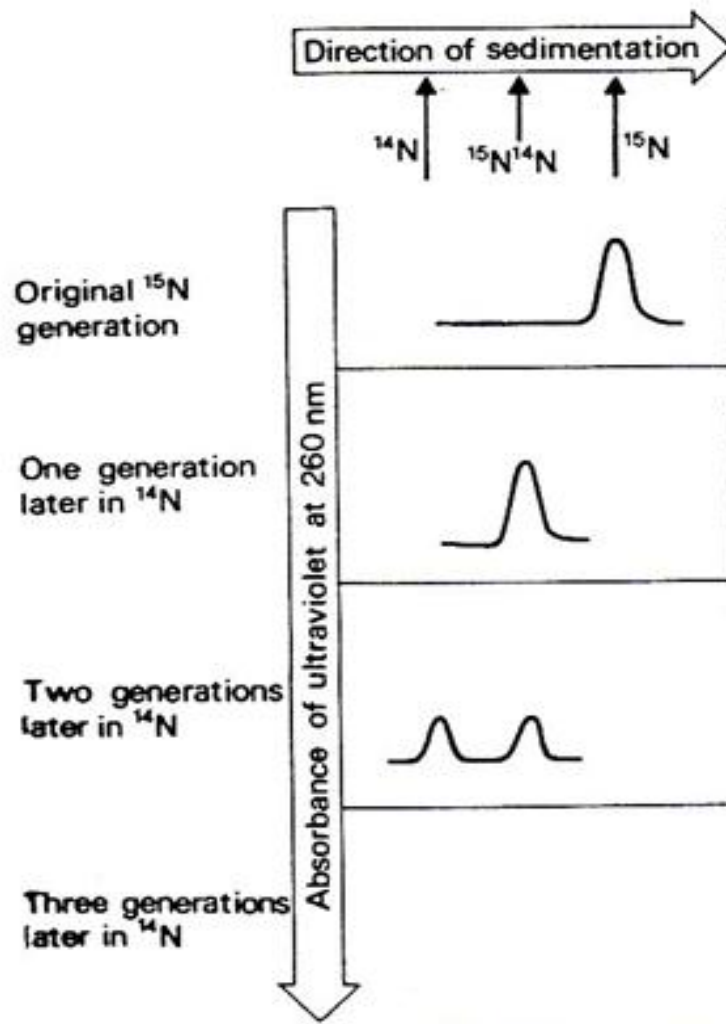


Fig. 2.1

- (a) (i) With reference to Fig. 2.1, do you consider the replication of DNA in *E. coli* as conservative or semi-conservative? Give reasons for your answer.

[3]

- (ii) On Fig. 2.1, draw the expected position of the ^{15}N and ^{14}N DNA absorbance after three generations in ^{14}N . [1]

- (iii) Explain the importance of hydrogen bonding in DNA structure.

[2]

- (iv) The enzyme that catalyses the replication of DNA checks for errors in the process and corrects them. Suggest why checking for errors and correcting them is necessary.

[1]

A group of scientists has discovered that a change in the chromatin structure of mammalian cell brought about by Protein A is associated with increased gene expression of several genes involved in the suppression of tumour growth. Fig. 2.2 shows the overall survival (OS) percentage of patients in the absence and presence of Protein A.

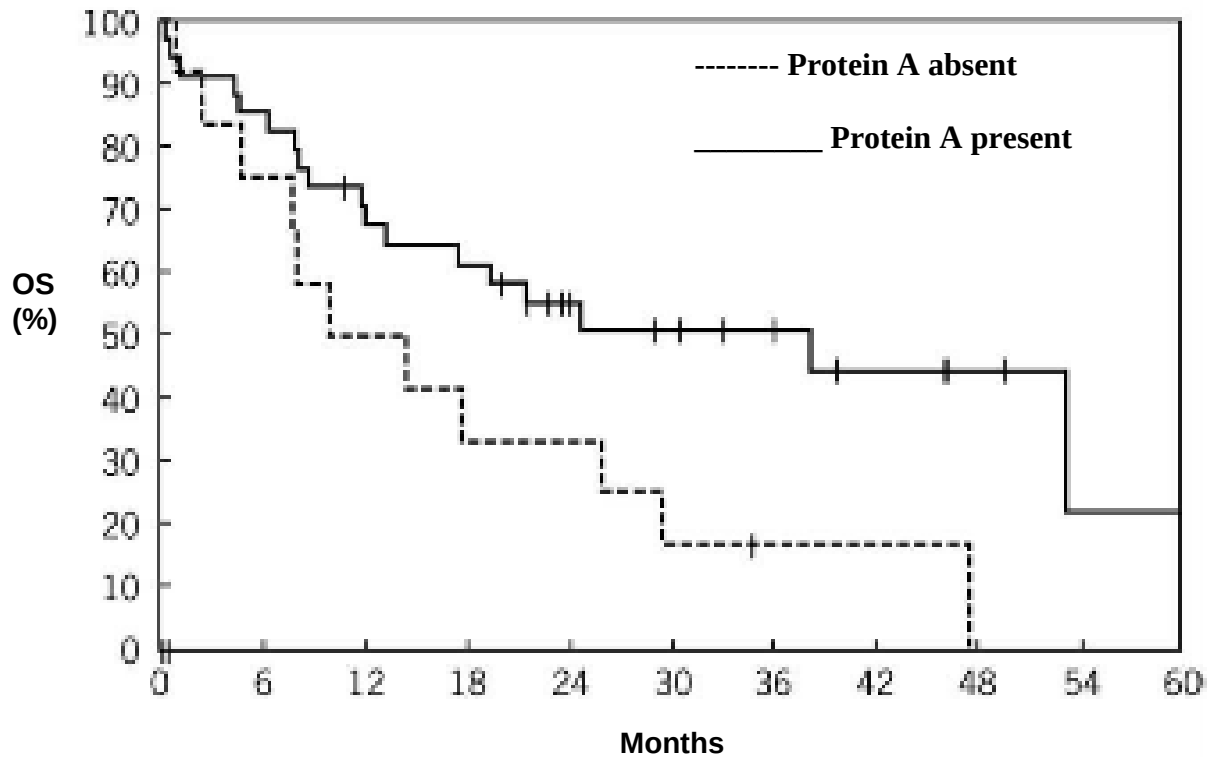


Fig. 2.2

- (b) (i) With reference to Fig. 2.2, comment on the overall survival percentage of patients in the presence of Protein A with that in the absence of Protein A, and explain its differences.

[4]

- (ii) Explain how the presence of transcription factors causes an increase in the expression of gene involved in suppression of tumour growth.

[3]

- (iii) Suggest why the change in chromatin structure, in the presence of Protein A, does not occur over large area of the chromatin.

[1]

[Total: 15]

- 3 Human Immunodeficiency Virus (HIV) is the culprit of Acquired Immunodeficiency Syndrome (AIDS) which, till now, has no full cure.

Figure 3.1 shows the reproductive cycle of HIV.

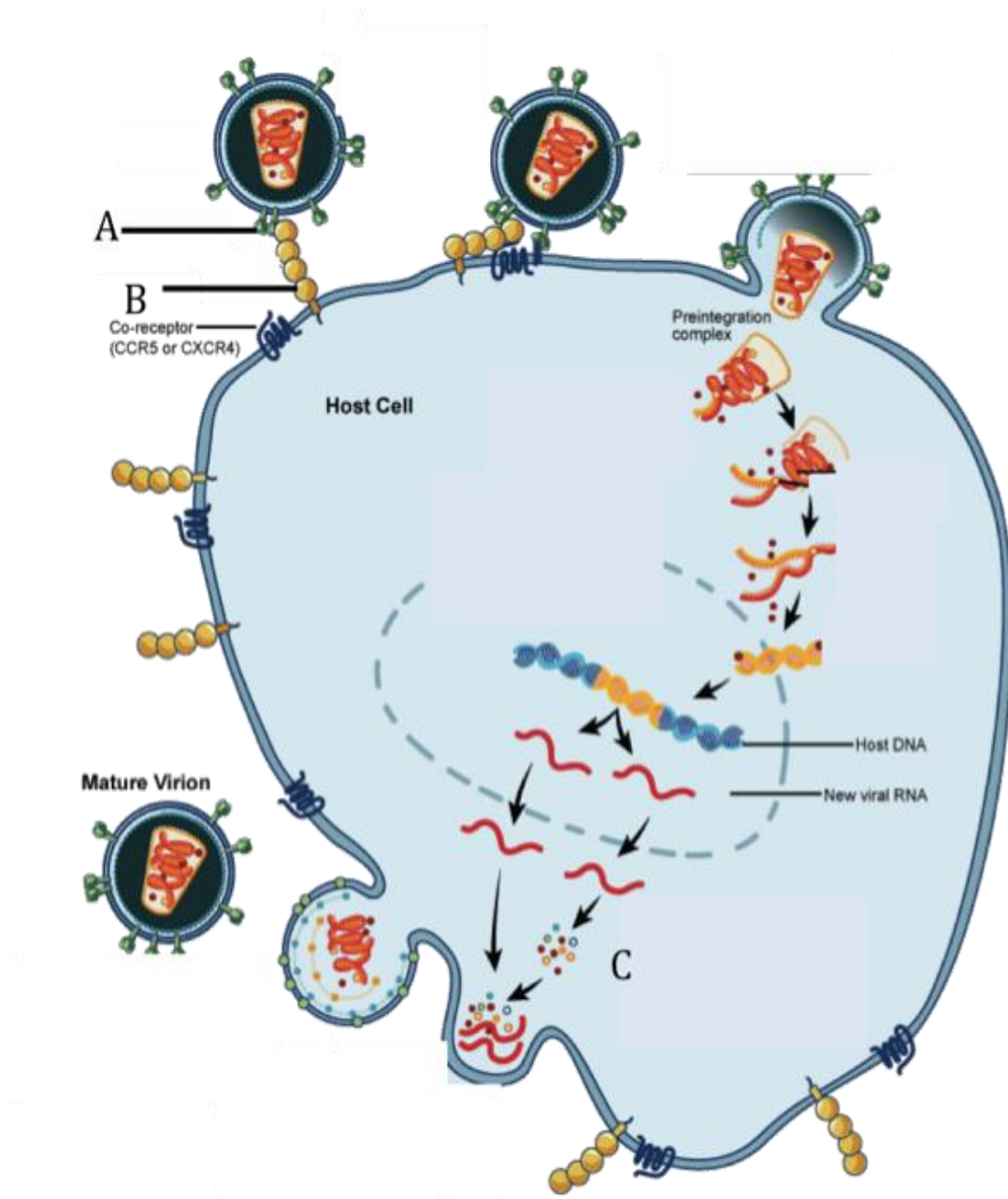


Fig.3.1

- (a) Name structure A and B.

Structure A:

Structure B:

[2]

- (b)** Describe the events occurring in stage C leading to formation of structure A on the host cell's plasma membrane.

[3]

- (c)** After contraction of HIV, the initial symptoms are followed by a stage called clinical latency when no symptoms are observed for a few years. With reference to the reproductive features of HIV, explain the cause of clinical latency.

[1]

- (d)** Flu is caused by influenza viruses.

- (i)** Distinguish the genome replication between HIV and influenza virus.

[2]

- (ii)** Every year, new flu vaccines are developed to ensure their efficacy. Explain the frequent need for new flu vaccines.

[2]

[Total: 10]

- 4 The arrow in Fig. 4.1 shows the direction of an impulse in part of a nervous system.

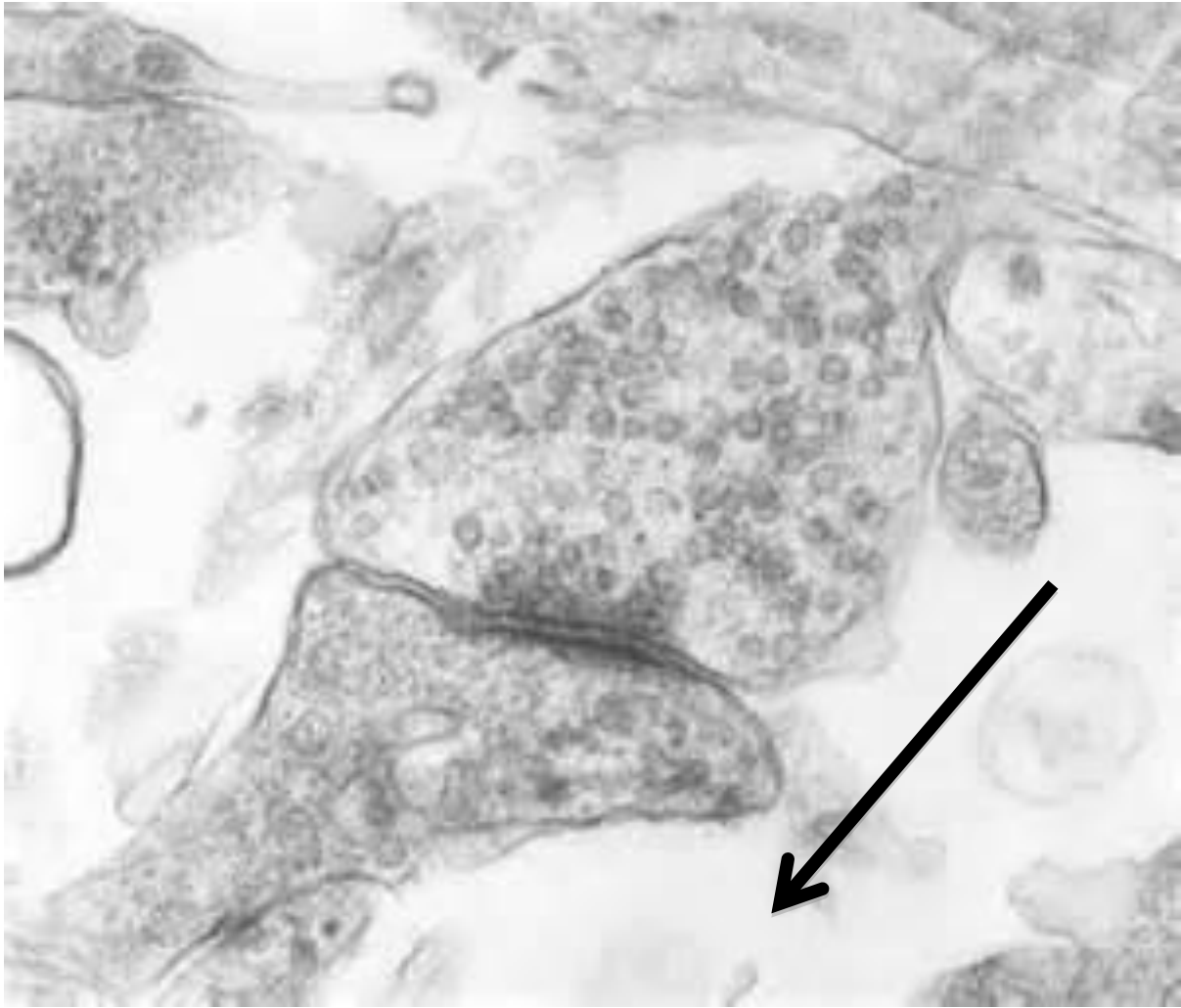


Fig. 4.1

- (a) With reference to Fig. 4.1, describe how transmission of impulse is believed to have taken place.

[4]

- (b) When summation of EPSP and IPSP exceeds threshold, depolarisation occurs. Explain why the magnitude of the signal initiated at the axon hillock is always at constant voltage.

[2]

- (c) Suggest the purpose of the refractory period during an action potential.

[2]

In a nerve muscle preparation, a novel drug called AWZ with a molecular structure resembling acetylcholine was applied to the cells. The investigation showed that the drug significantly increases muscular contractions. However, it was found that the drug, when applied to either one kind of cells directly, did not affect the axon nor muscle cells.

- (d) Suggest, in detail, one possible way in which the drug, AWZ, could cause the increased muscle contractions.

[4]

[Total: 12]

Stack 2

Name: _____

Class: _____

- 5 (a) Pituitary dwarfism is a sex-linked inherited condition in humans in which affected individuals have very short limbs. The family tree, in Fig. 5.1, shows part of one affected family.

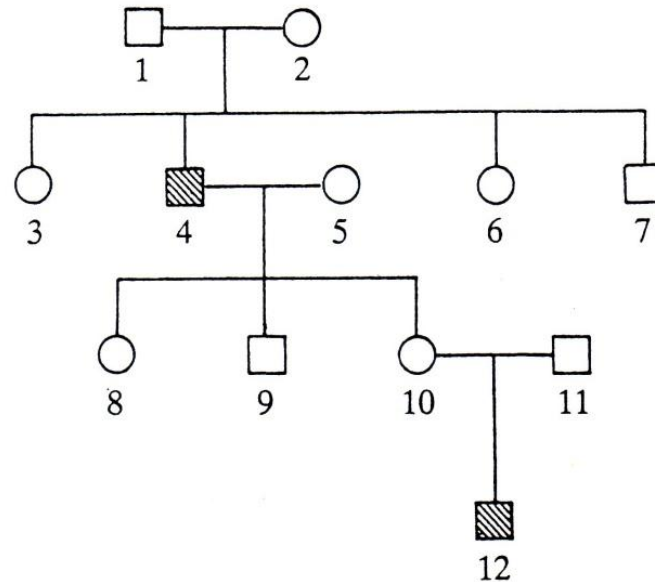


Fig. 5.1

- (i) Explain what is meant by 'sex-linked inherited condition'.

[1]

- (ii) With reference to Fig 5.1, explain the mode of inheritance using two pieces of evidences.

[4]

A series of breeding experiments were carried out with a certain species of mice. A cross between pure breeding males with red eyes and bent tails, and females with black eyes and straight tails produced offspring all with red eyes and bent tails. The same results were obtained when the cross was carried out again but with the sexes reversed.

- (b) Suggest a reason for carrying out the cross with the sexes reversed.

[2]

In another crossing, female offspring from the first experiment were then crossed with black eyes and straight tails males to obtain the following:

185 red eye, bent tail

153 red eye, straight tail

160 black eye, bent tail

188 purple eye, straight tail

- (c) Using the given formula and part of the table of chi-square values as shown in Table 5.1, determine if the null hypothesis for the eye color allele and tail shape allele can be accepted in the space below.

$$\chi^2 = \sum \frac{(o - e)^2}{e}$$

Part of the table of chi-square values is shown.

degrees of freedom	p = 0.5	p = 0.1	p = 0.05	p = 0.01	p = 0.001
1	0.46	2.71	3.84	6.64	10.83
2	1.39	4.6	5.99	9.21	13.82
3	2.37	6.25	7.82	11.34	16.27
4	3.36	7.78	9.49	13.28	18.46
5	4.35	9.24	11.07	15.09	20.52

Working to be shown in the space:

[3]

[Total: 10]

6 Fig. 6.1 illustrates what happens on thylakoid membrane during daytime.

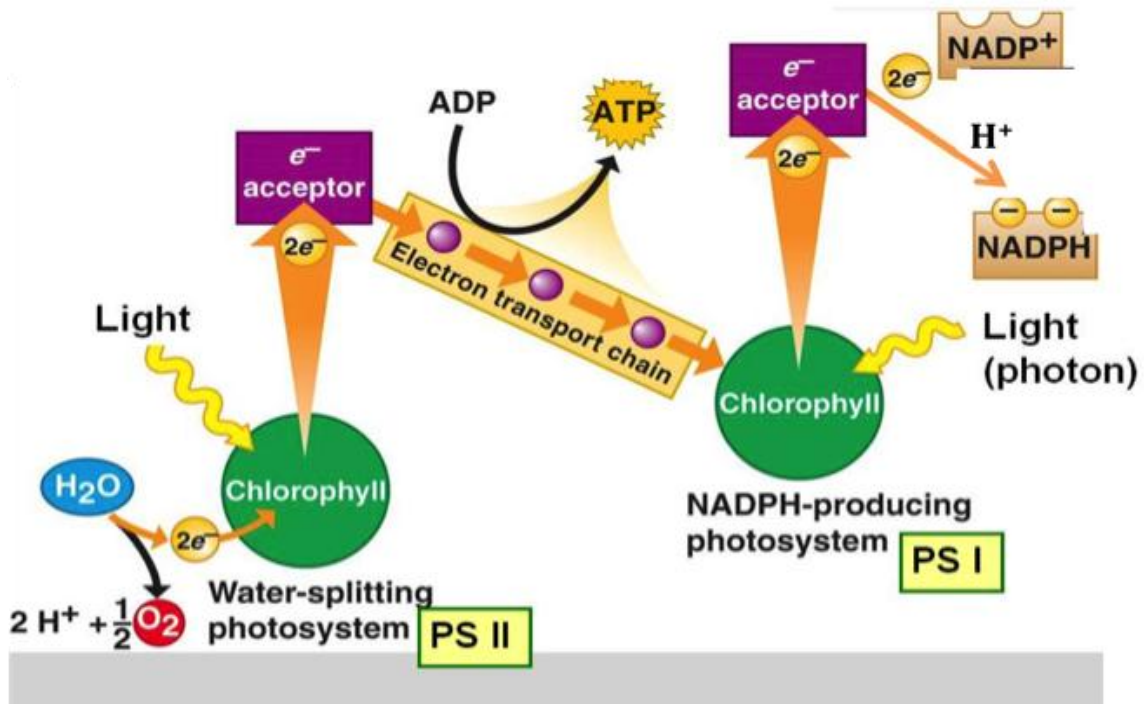


Fig.6.1

(a)

(i) Describe how electron transport chain leads to ATP production.

[3]

(ii) Compare the initial electron donor and final electron acceptor between oxidative phosphorylation and non-cyclic photophosphorylation.

[2]

- (b) A point mutation results in dysfunctional NADP⁺ reductase. State and explain how the concentration of Ribulose 1,5 – Bisphosphate in the chloroplasts would be affected.

[3]

- (c) Glyceraldehyde 3-phosphate exits the Calvin cycle for production of glucose, which will be used for cell wall formation. Explain how the structure of cellulose allows it to serve its function.

[4]

[Total: 12]

7 The greenish warbler, *Phylloscopus trochiloides*, is a species of small bird that originated in northern India, on the southern edge of the Himalayan mountain range.

(a) Describe the principles which scientists use to classify organisms into taxonomic groups.

[2]

Fig. 7.1 shows a greenish warbler.



Fig. 7.1

Thousands of years ago, populations of the greenish warbler spread around the western and eastern edges of the Himalayan mountain range to establish themselves in north-eastern Europe and Siberia.

- A gradual change in characteristics occurred in these populations, leading to different forms of the greenish warbler.
- One example of gradual change is in the song of the male warbler, which is very distinctive and is used in mating behavior.
- When greenish warblers from north-eastern Europe meet those from Siberia, no mating takes place.
- The greenish warblers from north-eastern Europe and Siberia are now considered to be two separate species.

Fig. 7.2 shows the spread of the greenish warbler.

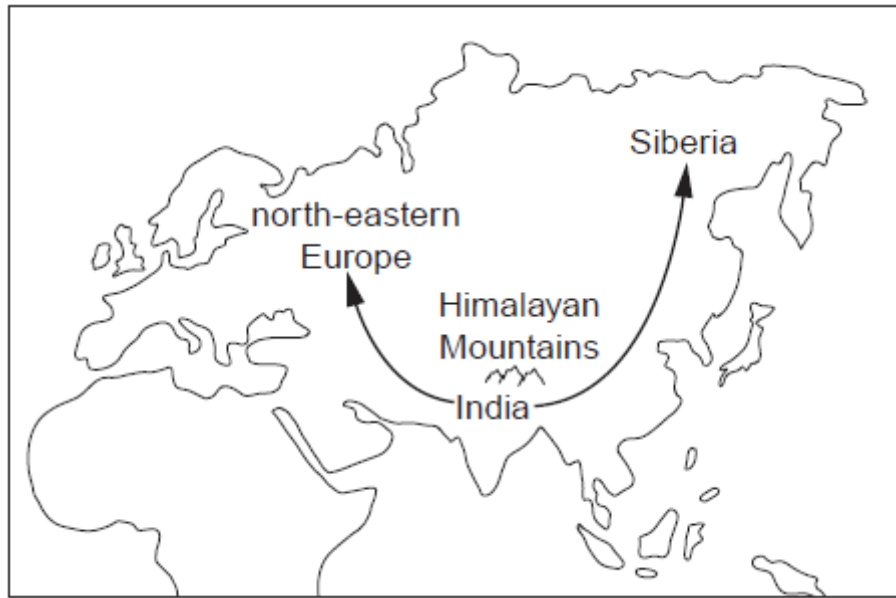


Fig. 7.2

- (b) State the likely isolating mechanism taking place in populations of the greenish warbler.

[1]

- (c) Explain how the process of speciation occurred in the greenish warbler populations.

[4]

An investigation was carried out to study the relationships between the different species of warbler using DNA analysis. The base sequences of a region of mitochondrial DNA from the four species were compared and results are shown in Table 7.1.

Table 7.1

	<i>P. laetus</i>			
<i>P. laetus</i>		<i>P. laurae</i>		
<i>P. laurae</i>	12.1		<i>P. ruficapilla</i>	
<i>P. ruficapilla</i>	16.7	15.0		<i>P. affinis</i>
<i>P. affinis</i>	11.3	8.9	13.2	

- (d) Researchers have put forward the hypothesis that the three species, *P. laetus*, *P. laurae* and *P. ruficapilla*, have originated from three separate events in which a few individuals of *P. affinis* spread directly from Cuba to three different places.

The smaller the number, the smaller the differences between the base sequences of the two species.

Explain how the results in Table 7.1 support the researchers' hypothesis.

[2]

- (e) It was observed that the warblers have remained at approximately the same frequency for many generations. Suggest an explanation for this phenomenon.

[1]

[Total: 10]

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 8** **(a)** Describe homologous chromosomes and distinguish between their behaviors in mitosis and the first division of meiosis. [10]
- (b)** With reference to specific genes, discuss how point mutations and chromosomal mutations can lead to cancer. [10]
- 9** **(a)** Describe how the effects of glucagon are achieved in the liver cell. [10]
- (b)** Discuss why the eukaryotic genome is considered to be more complex than the prokaryotic genome. [10]

--- End of Paper ---

